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PROJECT REPORT

ON

An LIBRARY MANAGEMENT SYSTEM

SUBMITTED BY:
920 – Makwana Gautam H.
928 – Hirani Sanee G.
937 – Vala Vivek R.

GUIDED BY:
Dr.Damyanti Patel

Acknowledgment

I am also deeply thankful to my friends and classmates for their continuous support, thoughtful discussions, and motivation during the development of the **Library Management System** project. Their contribution has helped me in shaping this project successfully.

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1. Introduction of Project

- **Objective:**

  - To design and develop a **Library Management System** that helps in **tracking, managing, and issuing books** efficiently for both **admin (librarian)** and **student (user)** interfaces.

- **Key Features:**

  i. Admin Panel:
     - Dashboard with total books, issued books, registered students
     - Add/Update/Delete books
     - View issued/returned books
     - Manage student record

       more..

  ii. Student Panel:
     - Login/Signup
     - View available books
     - Request book issue
     - View issued history

- **Security & Validation:**

  - Form validation (JS)
  - Password hashing (Express.js)
  - Role-based access control

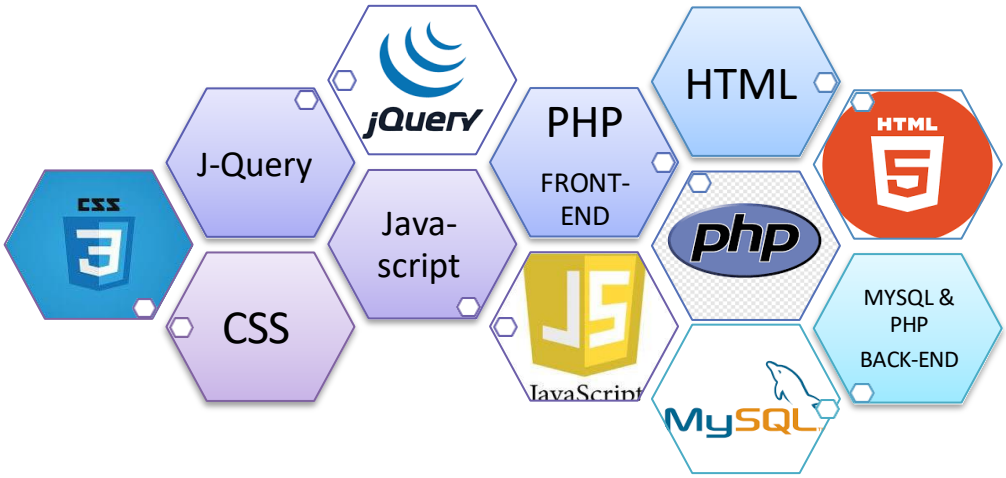
- **Project Status:**

  - Completed (CRUD operations, login, book issue/return)
  - Fully tested on Browser – localhost

2. Project Profile

Library Management System	
Project title:	Library Management System
Frontend:	HTML5, CSS, JavaScript ,Angular
Backend:	Express.js and MongoDB
Browser:	Chrome, Brave
Platform:	vs code
Documentation tool:	MS Word
Project Guide:	Dr. Damyanti Patel
Submitted to:	Dr. Damyanti Patel
Prepared By:	Gautam Makwana (Rno: 920), Sanee Hirani (Rno: 928), Vivek Vala (Rno: 937).

3. Technology Study



❖ **HTML,CSS,JS :**

- ALL are use to Frontend.
- JavaScript is a dynamic computer programming language.

❖ **Express.js,MongoDB:**

- This is used in backend side.
- Express.js code are executed on the server, and the result is returned to the browser as plain HTML.
- A database is a unstructured collection of data.

❖ **BOOSTRAP:**

- This is Framework for UI.
- Easy to use and responsive convert with grids.

Tools:

- **MS WORD:**
  - Microsoft Word or MS-WORD (often called Word) is a graphical word processing program that allows users to type and format text.
  - It is developed by Microsoft and is widely used for creating documents such as letters, reports, and resumes

- **Vs Code:**
  - VS Code is widely used for coding, debugging, and software development in various programming languages like Python, JavaScript, C++, Java, Express.js, and more.

  - Visual Studio Code (VS Code) is a free, open-source code editor developed by Microsoft.

4. Objectives

Objectives

- The main objective of the **Library Management System (LMS)** project is to design and develop an easy-to-use, responsive, and reliable web application that helps in managing books, journals, and study resources in a digital library.
- The system will provide features like browsing books, issuing and returning them, making reservations, keeping a wishlist, and giving feedback. Users will also be able to manage their profiles and check their history.
- For the librarian or admin, the system will include a secure panel to manage books, categories, users, and library transactions. This makes the overall process of handling library operations faster and more organized.

Start to End Process:

1. **Project Planning & Requirement Analysis**

- Define the purpose of the system: providing online access to books, journals, and academic resources.
- Identify key features for users and admins such as book browsing, search, reservation/wishlist, issue and return management, secure login, and admin book management.
- Finalize the technology stack:
  - Frontend: HTML, CSS, JavaScript, Bootstrap
  - Backend: Express.js (server-side scripting)
  - Database: MongoDB

2. **UI/UX Design**

- Design layouts for all major pages:
  - Homepage
  - Book Details Page
  - User Dashboard
  - Login/Register Page
  - Admin Panel
- Ensure a clean, professional, and responsive interface.
- Use a consistent theme, readable fonts, and neat layout to enhance usability.

3. **Frontend Development**

- **Use:**
  - HTML → structure
  - CSS → design & layout

- JavaScript → interactivity and validation

- Build pages such as:
  - Home (featured/new books)
  - Book Listing & Details
  - Login & Register
  - User Dashboard (issued books, edit Profile, profile)
  - Admin Panel UI

4. Backend Development

- **Use Express.js** to handle user sessions, book issue/return logic, and admin actions.
- **Connect to MongoDB** for managing backend data such as:
  - Users (students)
  - Books & Categories
  - Issue/Return Records
  - Feedback & Contact Messages
- Implement secure session management for both user and admin access.

5. Database Design

- Create essential tables:
  - Users (students/readers)
  - Admin
  - Books
  - Category
  - Author
  - Issued_Books
- Define relationships (e.g., Users → Issued_Books)

6. Functionality Implementation

User Side:

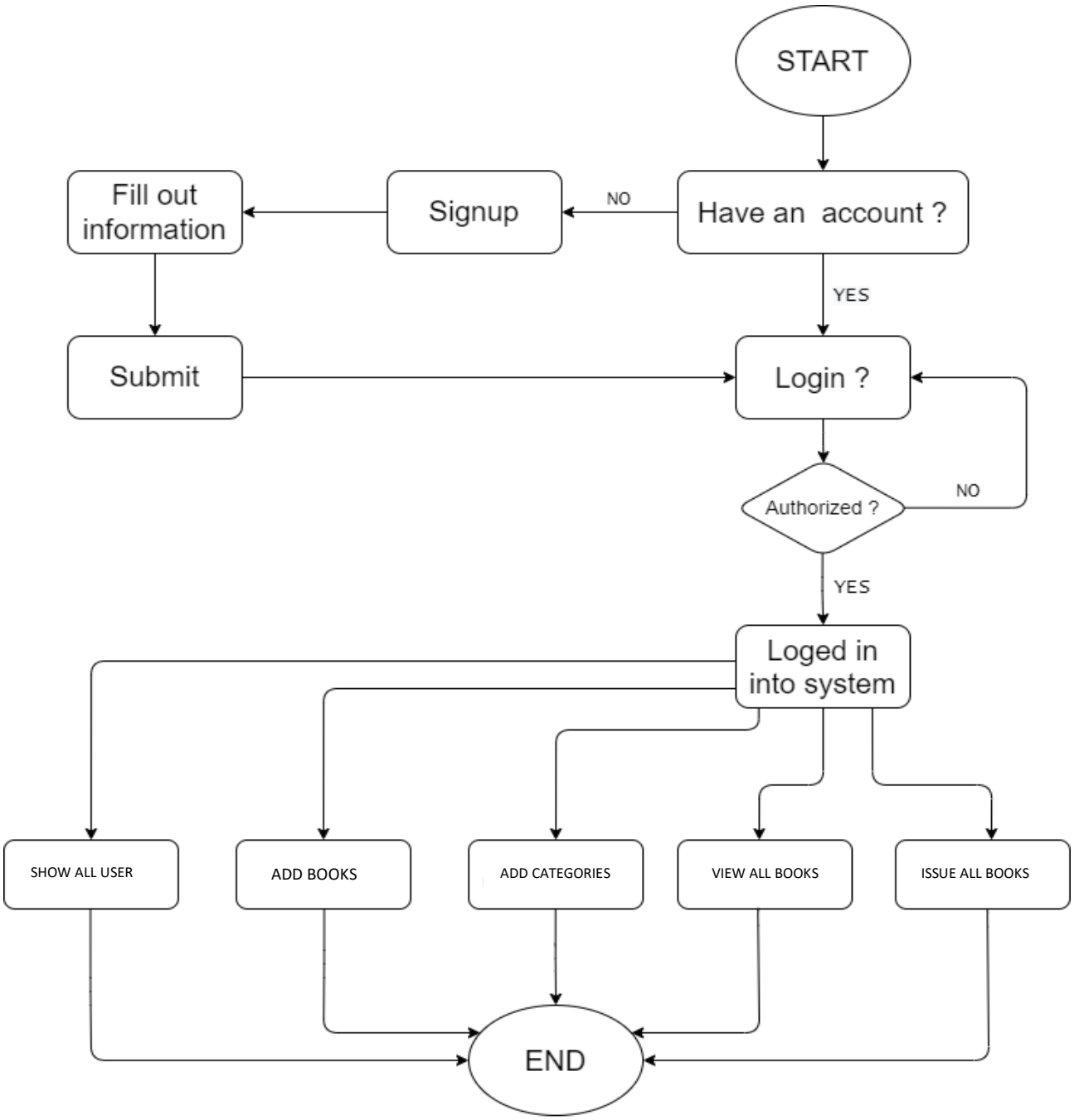
- User Registration & Login
- Browse and search for books by category/author
- Reserve or add books.
- Issue/Return books (based on rules)
- View profile,Edit Profile and changed password options

Admin Side:

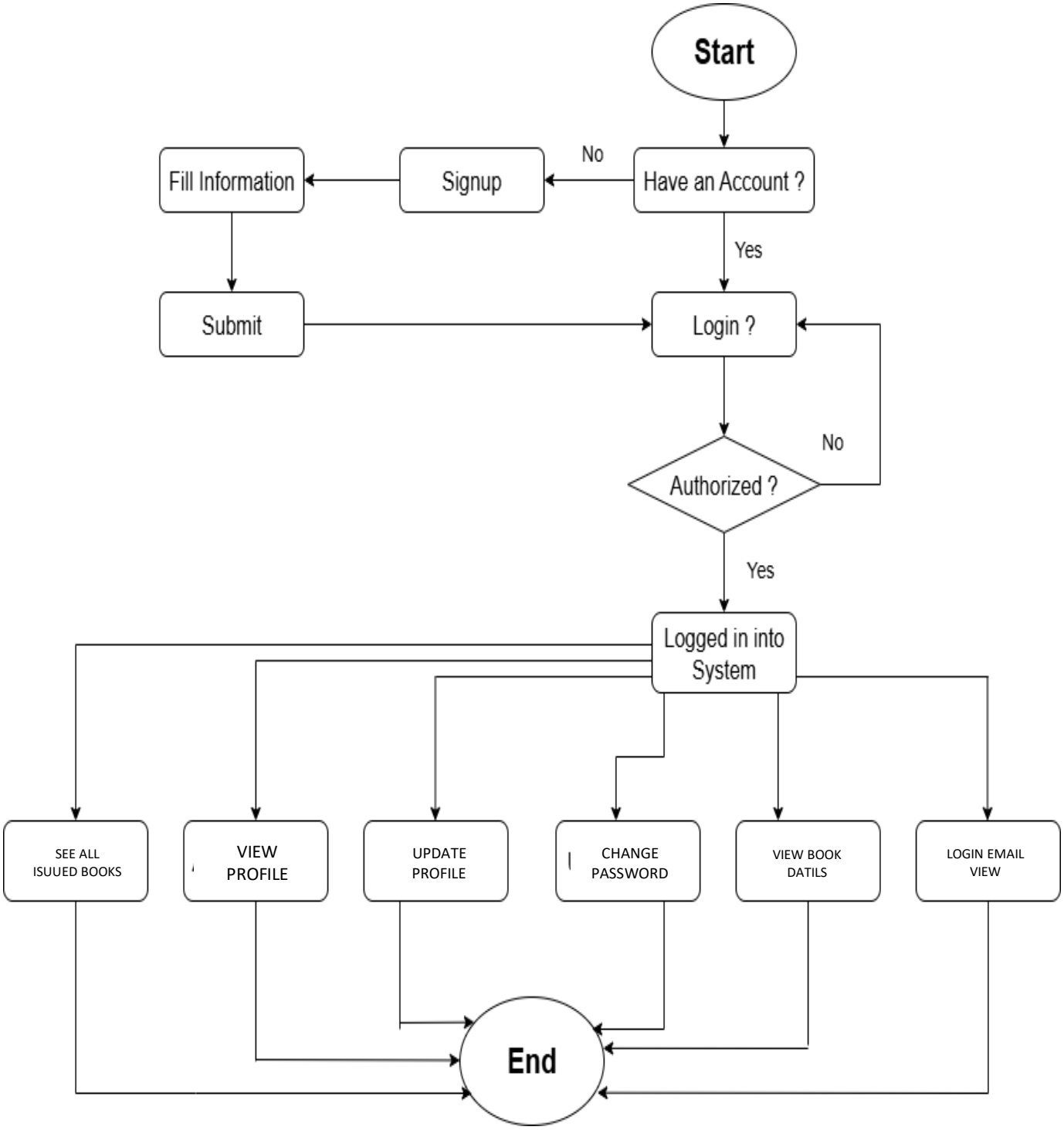
- Admin login with secure access
- Add/Edit/Delete books and categories
- Manage user accounts and transactions
- Track issued/returned books and overdue fines
- Dashboard overview of library activities (active users, most issued books, reservations, etc.

5. System Flow Diagram

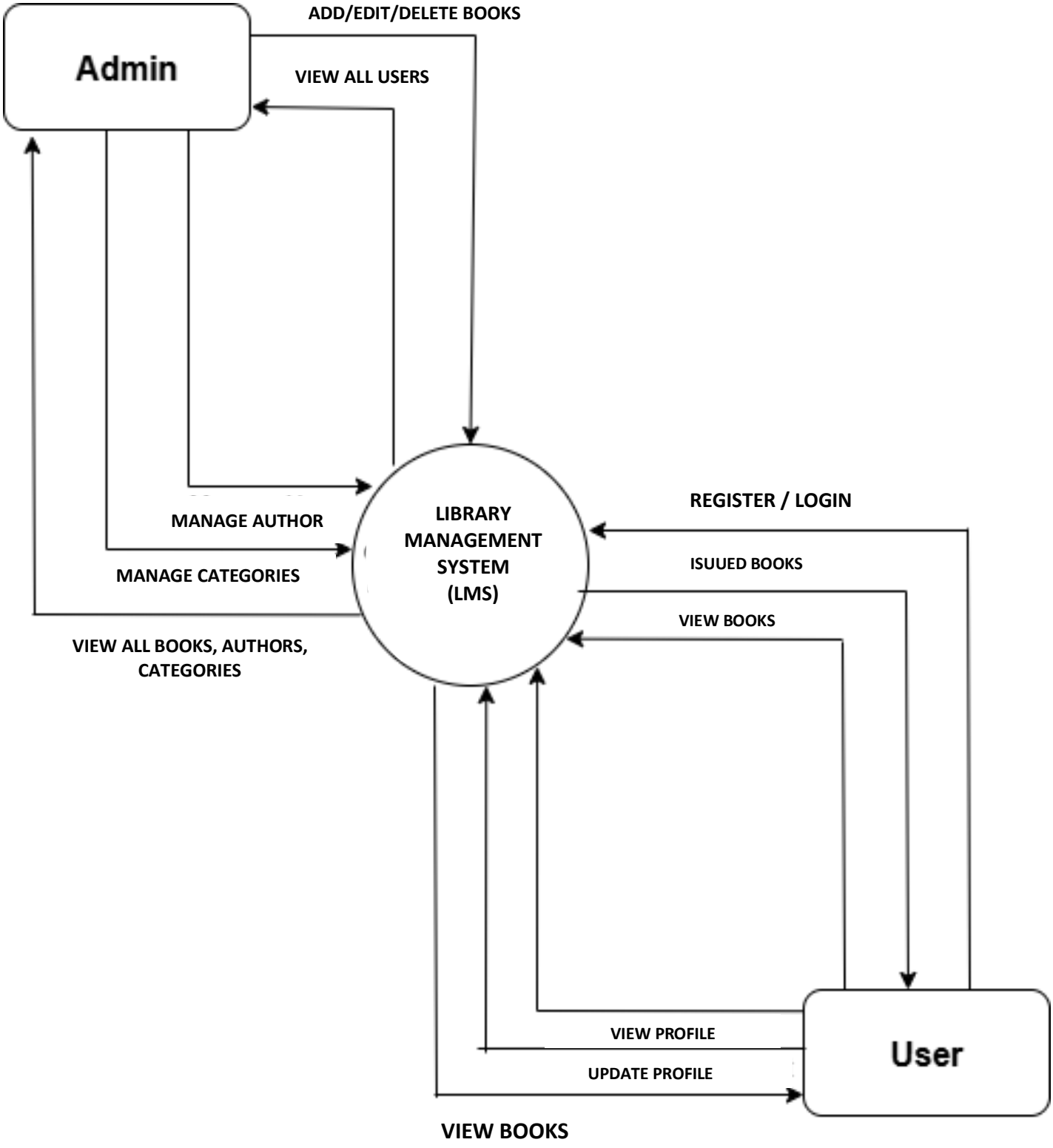
❖ Admin Process Flow Diagram:



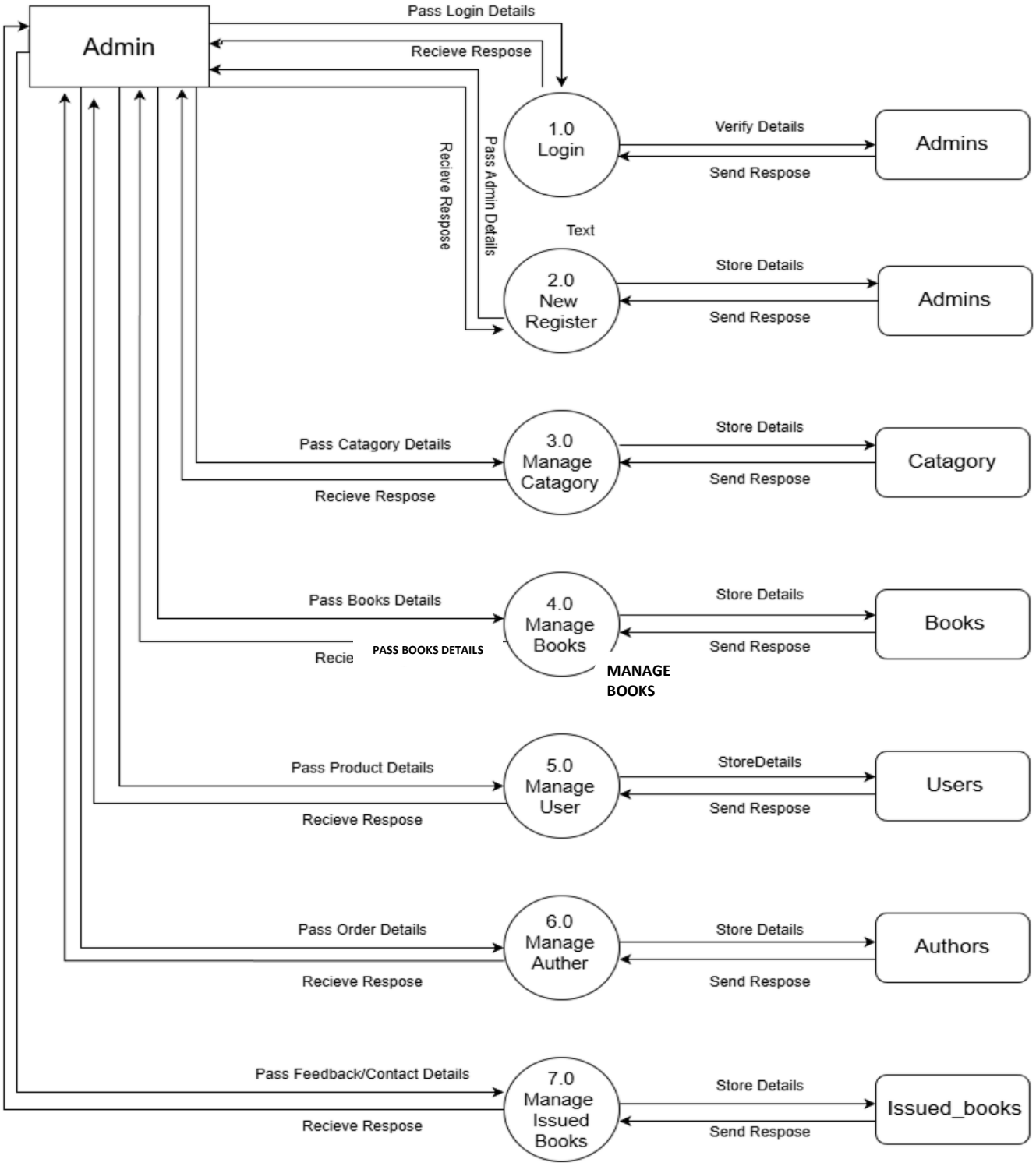
❖ User Process Flow Diagram:



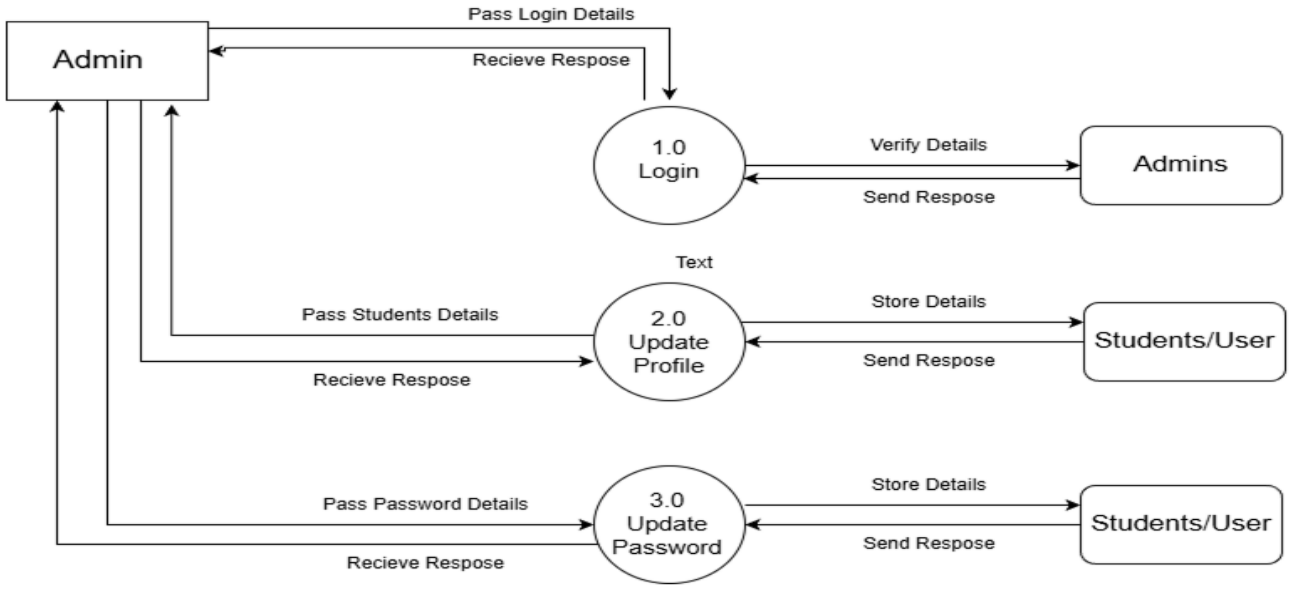
❖ 0 level Diagram(Context Diagram):



❖ 1st Level Admin Diagram:



❖ 1st Level User Diagram:



7. ER Diagram

id
email
password
name
Admin
mobile
Cat_id
id
name
email
Category
Users
Cat_name
MANAGE
mobile
address
password
Author_id
Authors
Author_name
S_no
Book_price
status
Book_id
Student_id
Isuued_books
BOOKS
Book_name
Book_autho
Isuued_date
Book_no
Cat_id
Author_id
Book_name
Book_no

9. Conclusion

The Library Management System project replaces the old manual work in libraries with a faster and easier computer-based system. It uses **Express.js** to run the backend and **MongoDB** to safely store all the data, such as book details, student information, and issue/return records. The website is built using **HTML, CSS, JavaScript, and Bootstrap , Angular**, which makes it look clean, modern, and simple to use.

The system helps librarians do important tasks more easily, like adding new books, issuing or returning books, registering students, and checking which books are available. This reduces manual work, avoids mistakes, and makes everyday library tasks quicker and more accurate.

In conclusion, this project shows how web technologies can be combined to build a useful and effective library system. It also gives a strong base for future improvements like barcode scanning, better reports, or allowing students to access the library online. This makes it a helpful tool for modern library needs

10.References

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- **Bootstrap Framework** – https://getbootstrap.com
- **GitHub** – https://github.com